Partial Derivatives and the Gradient

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: First and second partial derivatives

For $f(x, y) = e^{xy}\sin(x + y)$, find f_x using the product rule.

$f_x = ye^{xy}\sin(x + y) + e^{xy}\cos(x + y)$.

2 [Foundational] • Target: The gradient and directional derivatives

Find the gradient ∇f of $f(x, y) = x^2y + 3y^3$ at the point $(1, 2)$.

$\nabla f = \langle 2xy, x^2 + 9y^2\rangle$. At $(1, 2)$: $\nabla f(1, 2) = \langle 4, 37\rangle$.

3 [Foundational] • Target: The gradient and directional derivatives

Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4\rangle$.

$\nabla f = \langle 2x, 2y\rangle$, so $\nabla f(1, 1) = \langle 2, 2\rangle$. Unit vector: $\hat{\mathbf{v}} = \langle \frac{3}{5}, \frac{4}{5}\rangle$. $D_{\mathbf{v}}f = \langle 2, 2\rangle \cdot \langle \frac{3}{5}, \frac{4}{5}\rangle = \frac{6}{5} + \frac{8}{5} = \frac{14}{5}$.

4 [Foundational] • Target: Tangent planes

Find the equation of the tangent plane to $z = f(x, y) = x^2 + y^2$ at the point $(1, 2, 5)$, using $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

$f_x = 2x$, $f_y = 2y$: at $(1, 2)$, $f_x = 2$, $f_y = 4$. Tangent plane: $z - 5 = 2(x - 1) + 4(y - 2)$, i.e. $z = 2x + 4y - 5$.

5 [Foundational] • Target: Tangent planes

Find the equation of the tangent plane to $z = xy$ at the point $(2, 3, 6)$.

$f_x = y$, $f_y = x$: at $(2, 3)$, $f_x = 3$, $f_y = 2$. Tangent plane: $z - 6 = 3(x - 2) + 2(y - 3)$, i.e. $z = 3x + 2y - 6$.

6 [Foundational] • Target: Critical points of multivariable functions

Find the critical point(s) of $f(x, y) = x^2 + y^2 - 4x + 2y + 5$ by setting $f_x = 0$ and $f_y = 0$.

$f_x = 2x - 4 = 0$ gives $x = 2$; $f_y = 2y + 2 = 0$ gives $y = -1$. Critical point: $(2, -1)$.

7 [Foundational] • Target: Critical points of multivariable functions

Verify that the critical point found above is a minimum, using the second derivative test with $D = f_{xx}f_{yy} - (f_{xy})^2$.

$f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 0$: $D = 2(2) - 0 = 4 > 0$ and $f_{xx} = 2 > 0$, so $(2, -1)$ is a local minimum.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: The chain rule for multivariable functions**

Let $z = x^2y$, where $x = t^2$ and $y = t^3$. Use the multivariable chain rule $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ to find $\frac{dz}{dt}$ at $t = 1$.

$\frac{\partial z}{\partial x} = 2xy$, $\frac{\partial z}{\partial y} = x^2$; $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2$. At $t = 1$: $x = 1, y = 1$. $\frac{dz}{dt} = 2(1)(1)(2) + 1(3) = 4 + 3 = 7$.

9 **[Intermediate]** • **Target: The chain rule for multivariable functions**

Let $w = x^2 + y^2 + z^2$, where $x = \cos t$, $y = \sin t$, $z = t$. Find $\frac{dw}{dt}$ using the chain rule.

$\frac{\partial w}{\partial x} = 2x$, $\frac{\partial w}{\partial y} = 2y$, $\frac{\partial w}{\partial z} = 2z$; $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$, $\frac{dz}{dt} = 1$.

$\frac{dw}{dt} = 2x(-\sin t) + 2y\cos t + 2z = -2\cos t\sin t + 2\sin t\cos t + 2t = 2t$ (the first two terms cancel).

10 **[Intermediate]** • **Target: Maximum rate of increase**

Find the direction in which $f(x, y) = x^2 + xy$ increases most rapidly at the point $(1, 2)$, and state the maximum rate of increase.

$\nabla f = \langle 2x + y, x \rangle$; at $(1, 2)$: $\nabla f = \langle 4, 1 \rangle$. Direction of steepest increase: $\langle 4, 1 \rangle$ (or its unit vector). Maximum rate of increase: $|\nabla f| = \sqrt{16 + 1} = \sqrt{17}$.

11 **[Intermediate]** • **Target: Maximum rate of increase**

Find the direction of steepest descent for $f(x, y) = x^2 + y^2$ at the point $(3, 4)$.

$\nabla f = \langle 2x, 2y \rangle$; at $(3, 4)$: $\langle 6, 8 \rangle$. Steepest descent is the opposite direction: $\langle -6, -8 \rangle$ (pointing toward the origin, the minimum).

12 **[Intermediate]** • **Target: Implicit partial differentiation**

For the surface $x^2 + y^2 + z^2 = 9$, find $\frac{\partial z}{\partial x}$ by implicit differentiation (treating y as constant).

$2x + 2z\frac{\partial z}{\partial x} = 0$, so $\frac{\partial z}{\partial x} = -\frac{x}{z}$.

13 **[Intermediate]** • **Target: Classifying critical points with saddle points**

Find and classify the critical point(s) of $f(x, y) = x^2 - y^2$, using the second derivative test.

$f_x = 2x = 0$, $f_y = -2y = 0$ gives $(0, 0)$. $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$: $D = (2)(-2) - 0 = -4 < 0$, so $(0, 0)$ is a saddle point (neither a local max nor min).

14 **[Intermediate]** • **Target: Classifying critical points with saddle points**

Find and classify the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.

$f_x = 3x^2 - 3y = 0$ gives $y = x^2$. $f_y = -3x + 3y^2 = 0$ gives $x = y^2 = x^4$, so $x^4 - x = 0$, $x(x^3 - 1) = 0$: $x = 0$ or $x = 1$. Critical points $(0, 0)$ and $(1, 1)$. $f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = -3$. At $(0, 0)$: $D = 0 - 9 = -9 < 0$: saddle point. At $(1, 1)$: $D = (6)(6) - 9 = 27 > 0$ and $f_{xx} = 6 > 0$: local minimum.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: An applied optimization problem** • **Cross-Lesson: 8.02 Volumes of Revolution**

A company's profit is modeled by $P(x, y) = 100x + 120y - x^2 - 2y^2 - xy$, where x and y are units of two products. Find the production levels that maximize profit by setting $P_x = 0$ and $P_y = 0$.

$P_x = 100 - 2x - y = 0$; $P_y = 120 - 4y - x = 0$. From the first: $y = 100 - 2x$. Substituting: $120 - 4(100 - 2x) - x = 0$: $120 - 400 + 8x - x = 0$: $7x = 280$, $x = 40$. Then $y = 100 - 80 = 20$. Maximum profit at $(x, y) = (40, 20)$.

16 **[Challenging]** • **Target: Higher-order mixed partial derivatives** • **Cross-Lesson: 8.03 Arc Length and Improper Integrals**

For $f(x, y) = x^3y^2 + \sin(xy)$, find f_{yx} and f_{xy}, and verify they are equal (as guaranteed by Clairaut's theorem for continuous mixed partials).

$f_x = 3x^2y^2 + y\cos(xy)$: $f_{xy} = 6x^2y + \cos(xy) - xy\sin(xy)$. $f_y = 2x^3y + x\cos(xy)$: $f_{yx} = 6x^2y + \cos(xy) - xy\sin(xy)$. Both equal, confirming Clairaut's theorem.

17 **[Challenging]** • **Target: Linear approximation in two variables** • **Cross-Lesson: 9.01 Separable Differential Equations**

Use the linear approximation $f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$ to estimate $f(2.1, 2.9)$ for $f(x, y) = x^2y$, using the point $(2, 3)$.

$f(2, 3) = 12$. $f_x = 2xy$: $f_x(2, 3) = 12$. $f_y = x^2$: $f_y(2, 3) = 4$. Estimate: $12 + 12(0.1) + 4(-0.1) = 12 + 1.2 - 0.4 = 12.8$.

18 **[Challenging]** • **Target: Linear approximation in two variables** • **Cross-Lesson: 1.01 Precalculus Review for Calculus**

The volume of a cylinder is $V(r, h) = \pi r^2 h$. A cylindrical can is manufactured with radius $r = 3$ cm and height $h = 10$ cm, but due to manufacturing tolerances, the radius might be off by as much as 0.05 cm and the height by as much as 0.1 cm. Use the total differential $dV = V_r\,dr + V_h\,dh$ to estimate the maximum possible error in the computed volume.

$V_r = 2\pi rh$: at $(3, 10)$, $V_r = 60\pi$. $V_h = \pi r^2$: at $(3, 10)$, $V_h = 9\pi$. Maximum error estimate: $|dV| \le |V_r||dr| + |V_h||dh| = 60\pi(0.05) + 9\pi(0.1) = 3\pi + 0.9\pi = 3.9\pi \approx 12.25$ cm^3.

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19 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

For $f(x, y) = x^2y - 3xy^2 + y^3$, find each of the following at the point $(1, -1)$:

a $f_x(1, -1)$ $f_x = 2xy - 3y^2$: $f_x(1, -1) = 2(1)(-1) - 3(1) = -2 - 3 = -5$.

b $f_y(1, -1)$ $f_y = x^2 - 6xy + 3y^2$: $f_y(1, -1) = 1 - 6(1)(-1) + 3(1) = 1 + 6 + 3 = 10$.

c The gradient vector $\nabla f(1, -1)$ $\nabla f(1, -1) = \langle -5, 10 \rangle$.

d The directional derivative in the direction of $\langle 1, 0 \rangle$ $D_{\langle 1, 0 \rangle} f = \nabla f \cdot \langle 1, 0 \rangle = -5$ (since $\langle 1, 0 \rangle$ is already a unit vector).

(a) $f_x = 2xy - 3y^2$: $f_x(1, -1) = 2(1)(-1) - 3(1) = -2 - 3 = -5$.; (b) $f_y = x^2 - 6xy + 3y^2$: $f_y(1, -1) = 1 - 6(1)(-1) + 3(1) = 1 + 6 + 3 = 10$.; (c) $\nabla f(1, -1) = \langle -5, 10 \rangle$.; (d) $D_{\langle 1, 0 \rangle} f = \nabla f \cdot \langle 1, 0 \rangle = -5$ (since $\langle 1, 0 \rangle$ is already a unit vector).

20 [Challenging] • Target: An applied optimization problem • Cross-Lesson: 11.01 Sequences and Series

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.

Compute gradient: $ablaf = \langle 6xy, 3x^2 - 3y^2 \rangle$. At $(1, 2)$: $ablaf(1, 2) = \langle 6(1)(2), 3(1)^2 - 3(2)^2 \rangle = \langle 12, 3 - 12 \rangle = \langle 12, -9 \rangle$. Maximum rate of change is the gradient magnitude: $|ablaf(1, 2)| = \sqrt{12^2 + (-9)^2} = \sqrt{144 + 81} = \sqrt{225} = 15$ The direction of maximum increase is the normalized gradient: $\mathbf{u} = \frac{ablaf}{|ablaf|} = \frac{1}{15}\langle 12, -9 \rangle = \left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: First and second partial derivatives

Second-order Taylor approximation: $f(x, y) \approx f(a, b) + f_x(x - a) + f_y(y - b) + \frac{1}{2}[f_{xx}(x - a)^2 + 2f_{xy}(x - a)(y - b) + f_{yy}(y - b)^2]$. This involves:

a. only first partials

b. only second partials

c. third partials

d. **both first and second partials ✓**

Second-order Taylor uses both first and second partial derivatives.

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22 [Foundational] • Target: The gradient and directional derivatives

Chain rule with two paths: $z = f(x, y)$, $x = r\cos\theta$, $y = r\sin\theta$. $\partial z/\partial r =$

a. $f_x\cos\theta + f_y\sin\theta$ ✓

b. f_r

c. $f_x r + f_y r$

d. $f_x\sin\theta - f_y\cos\theta$

Chain rule: $z_r = f_x x_r + f_y y_r = f_x\cos\theta + f_y\sin\theta$.

23 [Foundational] • Target: Tangent planes

Classify the critical points of $f(x, y) = x^2 - 4x + y^3 - 12y$:

a. $(2, 2)$: saddle

b. $(2, 2)$ min; $(2, -2)$ saddle

c. $(2, 2)$**: local min;** $(2, -2)$**: local max** ✓

d. $(2, 2)$ saddle; $(2, -2)$ saddle

$f_x = 2x - 4 = 0 \Rightarrow x = 2$; $f_y = 3y^2 - 12 = 0 \Rightarrow y = \pm 2$. $H = (2, 6y, 0)$: at $(2, 2)$: $D = 2 \cdot 12 > 0$, $f_{xx} = 2 > 0 \Rightarrow$ min; at $(2, -2)$: $D = 2 \cdot (-12) < 0 \Rightarrow$ saddle.

24 [Foundational] • Target: Critical points of multivariable functions

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$. Name of λ:

a. **Lagrange multiplier** ✓

b. eigenvalue

c. gradient

d. curvature

Applying Application of Derivatives principles, λ = Lagrange multiplier. Therefore, the verified correct option is Lagrange multiplier.

25 [Foundational] • Target: The chain rule for multivariable functions

Second mixed partial: for smooth f, $f_{xy} =$

a. f_{yx} ✓

b. f_{yy}

c. f_{xx}

d. 0

Applying Partial Derivatives and Gradient principles, Clairaut: $f_{xy} = f_{yx}$. Therefore, the verified correct option is f_{yx}.

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26 [Foundational] • Target: Maximum rate of increase

Local extrema of f can occur only at critical points (interior) or on the:

a. axes

b. level curves

c. **boundary ✓**

d. Hessian eigenvalues

Global extrema on closed bounded region: check interior critical points and boundary.

27 [Foundational] • Target: Implicit partial differentiation

Find and classify all critical points of $f(x, y) = x^3 - 3x + y^3 - 3y$:

a. $(1, 1)$ **and** $(-1, -1)$ **minima;** $(1, -1)$ **and** $(-1, 1)$ **saddles ✓**

b. $(\pm 1, \pm 1)$: all minima

c. $(0, 0)$ only

d. no critical points

$f_x = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$; similarly $y = \pm 1$; check D at each.

28 [Foundational] • Target: Classifying critical points with saddle points

Tangent plane at $(2, 1)$ for $f = x^2y^3$: $f_x|_{(2,\,1)} = 4$, $f_y = 12$, $f(2, 1) = 4$. Approximation:

a. $4 + 4(x - 2) + 6(y - 1)$

b. $4 + 4(x - 2) + 12(y - 1)$ **✓**

c. $8 + 4(x - 2)$

d. $4 + 8(x - 2) + 12(y - 1)$

Linear approximation: $f \approx 4 + 4(x - 2) + 12(y - 1)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: An applied optimization problem

The second derivative test fails ($D = 0$). This means the critical point is:

a. always a saddle

b. **indeterminate without further analysis ✓**

c. always min

d. always max

$D = 0$: inconclusive — need higher-order analysis.

30 [Intermediate] • Target: Higher-order mixed partial derivatives

Implicit differentiation: $F(x, y) = 0$. Then $dy/dx =$

a. F_y/F_x

b. $-F_x/F_y$ ✓

c. F_x/F_y

d. $-F_y/F_x$

Implicit function theorem: $dy/dx = -F_x/F_y$.

31 [Intermediate] • Target: Linear approximation in two variables

Second-order Taylor for $f(x, y) = e^{x+y}$ at $(0, 0)$ up to second order:

a. $e(1 + x + y)$

b. $1 + x + y$

c. $1 + (x + y)^2/2$

d. $1 + x + y + \frac{1}{2}(x + y)^2$ ✓

$e^{x+y} \approx 1 + (x + y) + \frac{1}{2}(x + y)^2$.

32 [Intermediate] • Target: First and second partial derivatives

Linear approximation: $f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$. This is accurate near:

a. (a, b) **if** f **is differentiable ✓**

b. far from (a, b)

c. only when f is quadratic

d. when $ablaf = 0$

Linearisation is accurate near the point.

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33 [Intermediate] • Target: The gradient and directional derivatives

If $D < 0$, the critical point is a:

a. local min

b. local max

c. **saddle point** ✓

d. degenerate

Applying Partial Derivatives and Gradient principles, $D < 0 \Rightarrow$ saddle point. Therefore, the verified correct option is saddle point.

34 [Intermediate] • Target: Tangent planes

Find the maximum of $f(x, y) = xy$ on the ellipse $\frac{x^2}{4} + \frac{y^2}{25} = 1$.

a. 5 ✓

b. 10

c. $\frac{5}{2}$

d. $\frac{7}{2}$

$ablaf = \lambda ablag$ gives $y = \lambda \frac{2x}{4}$, $x = \lambda \frac{2y}{25}$, so $\frac{x^2}{4} = \frac{y^2}{25} = \frac{1}{2}$. Then $xy = 5$.

35 [Intermediate] • Target: Critical points of multivariable functions

Implicit function theorem: at (a, b), $\partial y/\partial x =$

a. $D_x F$

b. $-(D_y F)^{-1} D_x F$ ✓

c. 0

d. I

Applying Partial Derivatives and Gradient principles, IFT derivative formula. Therefore, the verified correct option is $-(D_y F)^{-1} D_x F$.

36 [Intermediate] • Target: The chain rule for multivariable functions

$f(x, y) = xy$: at origin, the second derivative test gives $D = f_{xx} f_{yy} - f_{xy}^2 = 0 - 1 = -1 < 0 \Rightarrow$

a. **saddle point** ✓

b. local max

c. local min

d. inconclusive

Applying Partial Derivatives and Gradient principles, $D < 0 \Rightarrow$ saddle. Therefore, the verified correct option is saddle point.

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37 [Intermediate] • Target: Maximum rate of increase

Euler's theorem for homogeneous functions: if $f(tx, ty) = t^n f(x, y)$, then $xf_x + yf_y =$

a. f

b. nf ✓

c. $n(n-1)f$

d. 0

Applying Partial Derivatives and Gradient principles, Euler: $xf_x + yf_y = nf$. Therefore, the verified correct option is nf.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Implicit partial differentiation • Cross-Lesson: 1.01 Precalculus Review for Calculus

For $f(x, y) = 4x^2 + 3y^2 - 8x - 6y + 4$: critical point and value:

a. $(1, 2)$ with $f = -1$

b. $(1, 1)$ with $f = 0$

c. $(2, 1)$ with $f = 0$

d. $(1, 1)$ **with** $f = -3$ ✓

Setting $f_x = 8x - 8 = 0$ and $f_y = 6y - 6 = 0$ gives $(x, y) = (1, 1)$, where $f(1, 1) = 4 + 3 - 8 - 6 + 4 = -3$.

39 [Challenging] • Target: Classifying critical points with saddle points • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Find the maximum of $f(x, y) = 5x + 12y$ on the circle $x^2 + y^2 = 25$.

a. 85

b. 325

c. 65 ✓

d. 169

The maximum occurs where $ablaf = \lambda ablag$, i.e. at $(x, y) = \frac{5}{\sqrt{169}}(5, 12)$. Then $f = 65$.

40 [Challenging] • Target: An applied optimization problem • Cross-Lesson: 11.01 Sequences and Series

Find the tangent plane to the surface $x^2 + y^2 - z = 4$ at $(1, 2, 1)$:

a. $2x + 4y + z = 9$

b. $x + 2y - z = 4$

c. $2x + 4y - z = 9$ ✓

d. $x + 2y = 5$

$F = x^2 + y^2 - z - 4$: $ablaF = (2, 4, -1)$ at $(1, 2, 1)$; plane: $2(x-1) + 4(y-2) - (z-1) = 0 \Rightarrow 2x + 4y - z = 9$.

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41 [Challenging] • Target: Higher-order mixed partial derivatives • Cross-Lesson: 12.01 Convergence Tests

Find the tangent plane to $z = \ln(x^2 + y^2)$ at the point where $(x,\ y) = (1,\ 0)$.

a. $z = 2x - 2$ ✓

b. $z = 2y$

c. $z = 2 - 2x$

d. $z = 2x$

$f_x = 2$, $f_y = 0$, $f = 0$ there. $z = f + f_x(x - x_0) + f_y(y - y_0)$ gives $z = 2x - 2$.

42 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 12.02 Power Series and Taylor Series

Classify the critical point $\left(-\frac{24}{47},\ \frac{3}{47}\right)$ of $f(x, y) = 3x^2 + xy + 3x + 4y^2$.

a. local maximum

b. saddle point

c. the test is inconclusive

d. **local minimum** ✓

$D = f_{xx}f_{yy} - f_{xy}^2 = 47$ at the point, and $f_{xx} = 6$, so it is a local minimum.

43 [Challenging] • Target: First and second partial derivatives • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Directional derivative $D_{\hat{u}}f = ablaf \cdot \hat{u}$. For $f = xy$, $\hat{u} = (1/\sqrt{2}, 1/\sqrt{2})$, at $(1, 1)$:

a. 2

b. 1

c. 0

d. $\sqrt{2}$ ✓

$ablaf = (y, x) = (1, 1)$; $(1, 1) \cdot (1/\sqrt{2}, 1/\sqrt{2}) = 2/\sqrt{2} = \sqrt{2}$.

44 [Challenging] • Target: The gradient and directional derivatives • Cross-Lesson: 15.01 Double Integrals

Find the maximum of $x + y + z$ on the sphere $x^2 + y^2 + z^2 = 3$.

a. 3 ✓

b. 9

c. $\sqrt{3}$

d. $3\sqrt{3}$

$ablaf = \lambda ablag$ gives $x = y = z$, so $3x^2 = 3$ and the maximum is $3x = 3$.

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45 [Challenging] • Target: Tangent planes • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Maximum of xy on ellipse $x^2/4 + y^2/9 = 1$ by Lagrange:

a. $3/2$ ✓

b. 3

c. $9/4$

d. 6

Setting up Lagrange: $y = \lambda x/2$, $x = 2\lambda y/9 \Rightarrow xy = 3/2$.

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